

Uni-VLaT: Whole-Body Tactile Adaptation of VLA Policies for Humanoid Loco-Manipulation

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Fig. 1. Uni-VLaT equips a humanoid with distributed whole-body touch for contact-rich interaction. Using temporal tactile context and training-time prediction of future tactile, proprioceptive, and visual representations, Uni-VLaT targets table sweeping, box holding, back-tap walking, human-robot hugging, and composed cleanup.

Abstract—Pretrained vision-language-action (VLA) policies provide reusable visual, linguistic, and action priors, but contact-rich whole-body control remains difficult when physical interaction is weakly observed or visually occluded. We present Uni-VLaT, a framework that adapts pretrained VLA policies with whole-body tactile sensing. A topology-aware encoder combines axial pooling with region-wise embeddings to process distributed tactile measurements, while a temporal module summarizes a causal four-frame history. The resulting tactile features interact with vision-language and robot-state features through the GR00T policy’s native attention pathways. A multimodal JEPA objective shapes a tactile-anchored cross-modal context through prediction of absolute future tactile, proprioceptive, and visual latents. Touch provides direct evidence of physical contact, while the three modality-specific targets describe complementary aspects of the ensuing interaction dynamics. Training-time predictive supervision shapes the tactile-conditioned policy used for whole-

body execution. Our evaluation focuses on five contact-rich behaviors on a Unitree G1 with a GR00T-based whole-body policy. **TODO: Insert the measured quantitative findings after completing all experiments.**

I. INTRODUCTION

Pretrained vision-language-action (VLA) models increasingly provide reusable visual, linguistic, and action priors across robot tasks and embodiments [1]–[7]. This capability is particularly valuable for humanoids, whose large workspaces and whole-body action spaces make task-specific policy learning expensive. Recent systems have extended large-scale or pretrained policies to humanoid loco-manipulation [8]–[11], yet task-level perception remains dominated by vision and proprioception. During sustained physical interaction, however,

vision and proprioception provide only indirect evidence of physical contact. Whole-body touch directly observes the body–environment interface, providing a crucial signal for contact-rich humanoid control.

Integrating tactile sensing into policy control is challenging because touch differs fundamentally from the modalities on which most pretrained policies are built. Tactile signals are sparse, local, and strongly dependent on physical contact and robot action, while visual and language features primarily describe scene semantics and task intent. Simply appending tactile features therefore does not guarantee that the policy will use them at the right stage of interaction or associate them with the corresponding body response and scene change. Moreover, introducing a new modality can disrupt representations learned by the pretrained policy if its role is not properly aligned with the existing multimodal and action-generation pathways. Effective tactile adaptation thus requires not only encoding touch, but also integrating it into the policy in a way that makes contact information meaningful for control.

Recent humanoid systems have begun to exploit distributed tactile sensing. TACT incorporates visual and tactile observations into whole-body imitation learning, WT-UMI uses tactile/force feedback for contact-aware planning and control, and HTD trains a dedicated multimodal Transformer with future tactile-latent and force prediction [12]–[14]. These works establish the value of whole-body touch, but leave two aspects underexplored in our setting. First, they largely rely on dedicated policy or planning architectures, leaving unclear how distributed touch can augment pretrained generalist robot policies while building on their existing semantic and action priors. Second, when predictive supervision is used, it is primarily centered on future tactile or force signals rather than jointly characterizing how contact evolves together with body state and visual observations.

Our key idea is to use touch as a *physical anchor* for cross-modal interaction modeling. Touch directly identifies physical interaction at the robot body, while proprioception captures the associated body response and vision captures changes in the surrounding scene. We therefore construct a tactile-anchored cross-modal context from contextualized tactile features and train it to predict future tactile, proprioceptive, and visual representations in their respective latent spaces. Rather than forcing these modalities into a single shared representation space, this formulation uses touch to connect their complementary observations of the same physical interaction.

We instantiate this idea in **Uni-VLaT**, a framework for adapting pretrained humanoid VLA policies with tactile sensing. Rather than treating touch as an additional input modality alone, Uni-VLaT uses tactile features as a physical anchor for multimodal interaction modeling. After interacting with visual, language, proprioceptive, and action representations inside the pretrained policy, the tactile features form a contact-centered cross-modal context. From this context, three modality-specific predictors estimate future tactile, proprioceptive, and visual representations. By jointly predicting how contact, body state, and scene appearance evolve, Uni-VLaT encourages the policy

to learn a representation of physical interaction that is directly relevant to control.

Our contributions are:

- **Touch-anchored cross-modal learning.** We introduce Uni-VLaT, a learning paradigm that uses tactile features as a physical anchor to connect tactile, proprioceptive, and visual observations through prediction of their future representations, encouraging the policy to capture the coupled evolution of physical interaction.
- **Pretrained VLA priors for tactile control.** We introduce a tactile adaptation strategy that leverages pretrained VLA semantic, visual, and action priors to ground touch in task context and support contact-rich control without retraining the policy from scratch.
- **Extensive humanoid evaluation.** We evaluate Uni-VLaT on five contact-rich Unitree G1 tasks under diverse interaction conditions, against multiple baselines, across three policy backbones, and with controlled ablations of the proposed tactile and predictive components.

II. RELATED WORK

a) Pretrained robot policies and tactile adaptation.:

Large-scale cross-embodiment datasets and pretrained generalist robot policies have enabled increasingly transferable vision-language-conditioned control across tasks and embodiments [1]–[7], [15]. Recent work has further explored how such pretrained policies can incorporate tactile and other non-visual sensing. Beyond Sight adapts generalist robot policies to heterogeneous vision, touch, and audio observations through language-grounded sensor finetuning [16]. Tactile-VLA connects tactile sensing to physical knowledge encoded in pretrained VLAs through tactile fusion and tactile-conditioned reasoning [17]. VLA-Touch augments a base VLA without finetuning through semantic tactile feedback for high-level planning and tactile-conditioned low-level action refinement [18]. TacCoRL integrates tactile feedback into pretrained VLA policies through sim-real co-training and simulation-based reinforcement learning [19]. HapticVLA transfers tactile-aware behavior into a VLA through tactile distillation, removing the need for tactile sensing at inference [20]. These approaches demonstrate that pretrained policy priors can support tactile-aware manipulation, but primarily focus on localized hand or end-effector interaction rather than distributed whole-body contact.

b) Humanoid whole-body tactile control.: Recent humanoid systems have advanced whole-body control and locomanipulation through large-scale policy learning [8]–[11], [21]–[23]. Despite this progress, tactile feedback remains uncommon in learned humanoid task policies. TACT incorporates upper-body tactile observations into whole-body imitation learning [12]. WT-UMI combines distributed tactile/force sensing with force-supervised planning and tactile-based admittance control [13]. HTD trains a dedicated multimodal Transformer with auxiliary prediction of future tactile latents and hand-joint forces for contact-rich humanoid manipulation [14]. These systems establish the value of distributed sensing for

whole-body physical interaction, but rely on dedicated tactile-control or planning architectures. In contrast, Uni-VLaT adapts pretrained generalist robot policies with distributed tactile sensing while retaining their existing semantic and action priors.

c) *Predictive representations for physical interaction.*

Predictive representation learning provides a complementary way to model interaction dynamics without reconstructing raw observations. I-JEPA and V-JEPA formulate joint-embedding prediction for images and video [24], [25], while V-JEPA 2 and DINO-WM extend latent prediction toward physical dynamics and action-conditioned planning [26], [27]. Predictive objectives have also been incorporated directly into robot policy learning: DreamVLA forecasts compact future world knowledge for action generation [28], Cosmos Policy jointly generates actions and future states for visuomotor control and planning [29], VLA-JEPA learns action-relevant representations through future latent-state prediction [30], and JEPA-WAM incorporates joint-embedding prediction into world-action modeling [31].

Recent work further introduces predictive tactile dynamics for contact-rich manipulation. Dream-Tac jointly models actions, future visual observations, and tactile dynamics [32]. Tactile-WAM predicts tactile contact dynamics while using asymmetric attention to preserve visual modeling and guide action generation [33]. VT-WAM jointly learns future visual, tactile-deformation, and action prediction within a unified world-action model [34]. TacWAM uses tactile history and mechanics-aware future tactile prediction as supervision for contact-aware action learning [35]. Together with HTD’s future tactile-latent and force prediction [14], these methods demonstrate the value of predictive tactile supervision, but primarily model tactile/force or visual–tactile futures. Uni-VLaT instead uses touch as a physical anchor to jointly supervise future tactile, proprioceptive, and visual representations from the tactile-conditioned policy context.

III. PROBLEM FORMULATION

We study language-conditioned humanoid loco-manipulation with distributed whole-body tactile feedback. At observation step t , the policy receives

$$\mathcal{O}_t = (I_t, \ell, q_t, \tau_{t-H_T+1:t}, \tau_{t-H_T+1:t} = (\tau_{t-3}, \dots, \tau_t), \quad (1)$$

where I_t denotes multi-view RGB images, ℓ a language instruction, q_t the proprioceptive state available to the policy, and τ_t the distributed tactile measurements. We use a causal history of $H_T = 4$ tactile frames.

Our action space is the 64-dimensional latent motion space of SONIC [9]. A pretrained VLA policy is adapted to predict a chunk of motion tokens, which are passed to SONIC’s whole-body control decoder:

$$\begin{aligned} A_t &= (a_{t,0}, \dots, a_{t,H_A-1}) \sim \pi_\theta(\cdot \mid \mathcal{O}_t), \\ q^{\text{des}} &= \mathcal{D}_{\text{SONIC}}(s, a), \quad a \in A_t. \end{aligned} \quad (2)$$

Here $A_t \in \mathbb{R}^{H_A \times 64}$ is a chunk of 64-dimensional latent motion tokens. At each control update, a denotes the token selected for execution from the active chunk, and s denotes SONIC’s controller observation, including joint positions and

velocities, base inertial measurements, and previous control outputs. The pretrained control decoder $\mathcal{D}_{\text{SONIC}}$ maps the selected motion token and robot feedback to desired joint positions q^{des} , which are tracked by joint-level PD controllers. We directly supply the predicted tokens to this decoder through SONIC’s Protocol v4 interface, enabling coordinated upper- and lower-body execution through a single whole-body controller.

Given synchronized demonstrations $\mathcal{D} = \{(\mathcal{O}_t, A_t^*)\}$, where A_t^* is the demonstrated motion-token chunk, our objective is to learn a tactile-conditioned VLA policy that generates task-appropriate whole-body motion from visual, linguistic, proprioceptive, and tactile observations. Learning adapts the high-level policy, while the pretrained SONIC controller serves as the low-level execution interface.

IV. METHOD

A. Method Overview

Uni-VLaT adapts a pretrained VLA policy through tactile integration and multimodal predictive supervision (Fig. 2). A spatial encoder and a temporal module transform distributed tactile measurements into tokens summarizing recent contact. These tokens interact with vision-language, proprioceptive, and action features within the policy’s DiT trunk. The action-token outputs generate motion commands, while the contextualized tactile-token outputs form a shared context for predicting future tactile, proprioceptive, and visual representations. Action supervision and the three predictive objectives jointly train the tactile-conditioned policy. The predictive heads and target encoders are used only during training.

B. Tactile Encoding and Policy Integration

Spatial tactile encoding. Our tactile sensors cover eight body regions: the chest, central back, left and right shoulders, left and right upper back, and left and right arms. We suppress sensor noise with a fixed deadband and normalize readings by region. For the tactile sleeves, we pool measurements around the arm circumference to reduce sensitivity to sleeve rotation while preserving contact location along the arm.

Let $\tau_{t,r}$ denote the tactile measurements for region r at time t after the preprocessing described above. Each region has an independent MLP E_r that maps its measurements to a common feature width, producing the regional embeddings $R_t = [E_r(\tau_{t,r})]_{r=1}^8$. To combine contact information across regions, we introduce a set of eight learned queries Q^T and use the regional embeddings as both keys and values:

$$S_t^T = \text{LN}(\text{CrossAttn}(Q^T, R_t, R_t)), \quad (3)$$

where the three attention arguments denote queries, keys, and values, respectively, and LN denotes layer normalization. The superscript T denotes the tactile modality. The output S_t^T contains eight tactile tokens, each of which can aggregate information from multiple body regions. The learned queries are shared across frames, providing a consistent token ordering for temporal encoding.

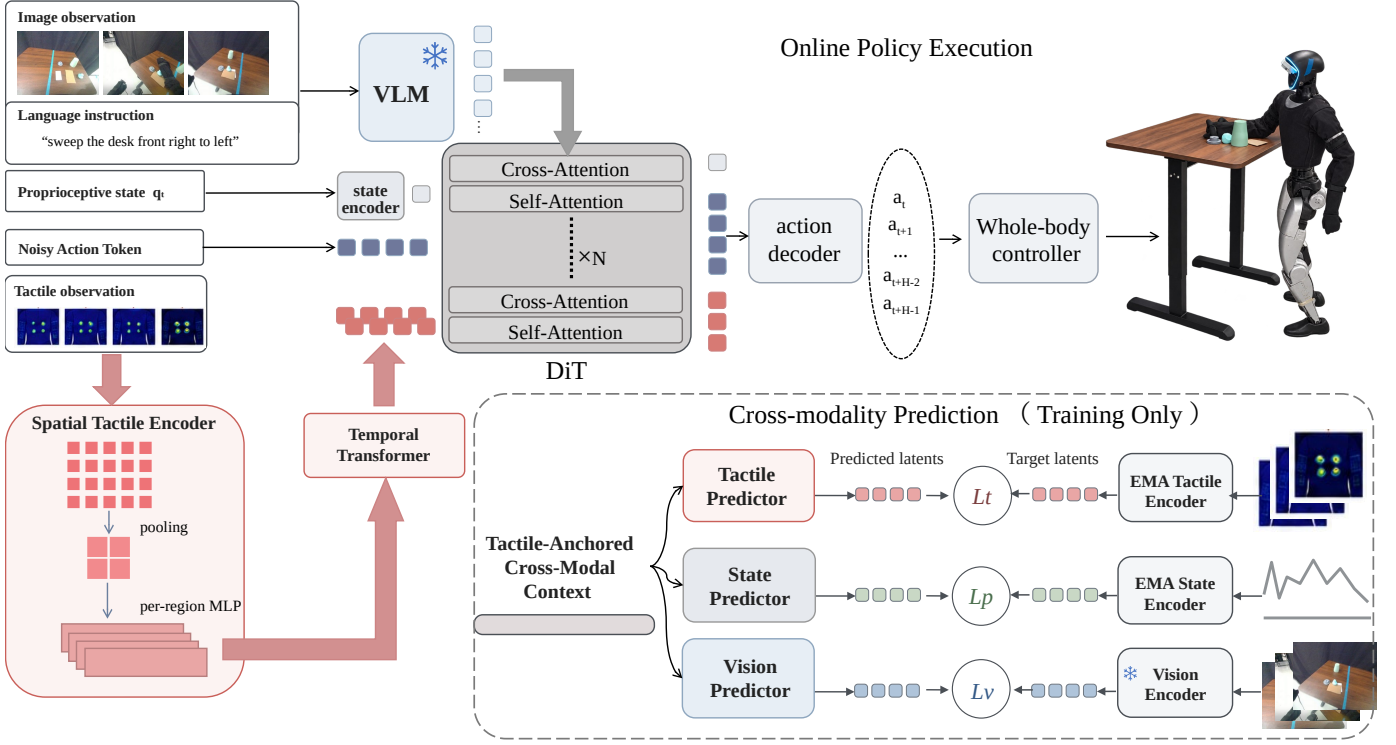


Fig. 2. Overview of Uni-VLaT. Spatial and temporal encoding transforms distributed tactile observations into tokens that interact with visual, language, proprioceptive, and action features within the pretrained policy. The action-token outputs support whole-body action generation, while the contextualized tactile tokens are pooled into a tactile-anchored cross-modal context. During training, three auxiliary branches predict future tactile, proprioceptive, and visual latents from this context, jointly supervising the policy with the action-learning objective.

Temporal contact encoding. A single tactile observation may be insufficient to distinguish contact establishment, sustained support, and release. We therefore use a lightweight temporal Transformer to aggregate the spatial tactile tokens over a four-frame history:

$$Z_t^T = \Phi_T(S_{t-3}^T, S_{t-2}^T, S_{t-1}^T, S_t^T). \quad (4)$$

Here Φ_T applies temporal attention to the four-frame sequence at each tactile-token index, using learned time embeddings to encode frame order. The aggregated features are combined with the current-frame tokens through a residual connection, producing temporally informed tactile tokens Z_t^T . A learned scalar gate scales these tokens before they enter the pretrained policy.

Integration into the action head. The VLA action head uses a DiT trunk to jointly process the gated tactile tokens, one proprioceptive token, and noisy action tokens. Visual and language features from the pretrained VLM are supplied through cross-attention. Self-attention enables interaction among tactile, proprioceptive, and action tokens, while cross-attention incorporates scene and instruction information. This contextualizes contact with the robot’s body state and task, while making tactile information available for action generation.

The action-token outputs are projected to flow-matching velocity predictions, which are used to generate the motion-token chunk through iterative integration. The tactile-token

outputs, denoted by $h_{t,1}^T, \dots, h_{t,8}^T$, are retained for the predictive supervision described next. Both outputs come from the same DiT trunk, linking tactile contextualization to the action-generation pathway.

C. Tactile-Anchored Multimodal Prediction

Predictive context. Through self-attention and cross-attention in the DiT trunk, tactile tokens incorporate information from proprioceptive, action, and vision-language features. The resulting post-DiT tactile representations therefore provide a contact-centered context for predicting how the interaction evolves across modalities. To obtain a compact input for the predictive heads, we average these contextualized tokens:

$$c_t = \frac{1}{8} \sum_{n=1}^8 h_{t,n}^T. \quad (5)$$

Here c_t is a compact readout of the *tactile-anchored cross-modal context* learned through policy attention. Mean pooling reduces the predictor input size. Three modality-specific MLP predictors map c_t to future latent sequences:

$$\hat{z}_{t+1:t+K}^m = P_m(c_t), \quad K = 4, \quad m \in \{T, P, V\}, \quad (6)$$

where T , P , and V denote tactile, proprioceptive, and visual modalities. Each predictor jointly outputs a sequence of four latents, one for each future observation step.

Future target representations. The prediction targets are encoded from the corresponding future observations:

$$\begin{aligned} z_{t+k}^T &= \text{sg}(\bar{E}_T(\tau_{t+k})), \\ z_{t+k}^P &= \text{sg}(\bar{E}_P(q_{t+k})), \\ z_{t+k}^V &= \text{sg}(\bar{E}_V(I_{t+k})), \end{aligned} \quad (7)$$

where $k = 1, \dots, K$ and sg denotes stop-gradient. For touch, the target encoder is an exponential-moving-average (EMA) copy of the online spatial tactile encoder, with decay 0.99. It encodes each future frame independently and averages its output tokens into one latent, while the online pathway additionally aggregates the four-frame history to provide predictive context. For proprioception, an EMA copy of the online state encoder, also with decay 0.99, embeds the future robot state. For vision, the frozen pretrained vision tower extracts patch features, which are averaged within each image and then across camera views. Each modality thus provides one target latent per future step.

All branches predict absolute future latents in their respective modality-specific spaces. These targets retain both persistent interaction states, such as sustained contact during box support, and changes in contact, posture, and scene appearance. Joint prediction encourages c_t to capture these complementary aspects of the interaction without forcing the three modalities into a common target space.

Joint objectives and gradient flow. Following HTD [14], we combine cosine-based directional alignment with SmoothL1 regression of latent magnitudes, applying this objective to each of the three predicted modalities:

$$\mathcal{L}_m = \frac{1}{K} \sum_{k=1}^K \left[1 - \cos(\hat{z}_{t+k}^m, z_{t+k}^m) + \beta \text{SmoothL1}(\|\hat{z}_{t+k}^m\|_2, \|z_{t+k}^m\|_2) \right], \quad (8)$$

where $\beta = 1$. The cosine term aligns feature directions, while the SmoothL1 term constrains feature magnitudes. The overall training objective is

$$\mathcal{L} = \mathcal{L}_{\text{act}} + \lambda_T \mathcal{L}_T + \lambda_P \mathcal{L}_P + \lambda_V \mathcal{L}_V, \quad (9)$$

where $\mathcal{L}_{\text{act}}$ is the flow-matching velocity MSE over valid action dimensions. We use $\lambda_T = 0.05$ and $\lambda_P = \lambda_V = 0.005$ across tasks.

Action-loss gradients propagate through the action-token outputs and the shared DiT attention pathways to the trainable tactile encoding modules. Auxiliary-loss gradients pass through the three predictors and c_t to the post-DiT tactile tokens, updating the predictors and the trainable policy and tactile modules along these paths. Thus, tactile features receive supervision both from demonstrated actions and from future multimodal observations. Stop-gradient prevents predictive losses from updating the target encoders directly; tactile and proprioceptive teachers instead follow their online encoders through EMA updates, while the visual teacher remains frozen.

Deployment. At deployment, a rolling buffer supplies the four-frame tactile history, which is encoded into tokens for the

tactile-conditioned VLA policy. The policy generates chunks of 64-dimensional motion tokens, and the token selected at each control update is sent to SONIC’s control decoder through its Protocol v4 interface. As defined in Eq. 2, the decoder combines this token with robot proprioceptive feedback to produce joint-position targets, which are tracked by joint-level PD controllers for coordinated upper- and lower-body execution. The predictive heads and target encoders are used only during training and are omitted at deployment.

V. EXPERIMENTS

Our experiments examine whether multimodal predictive supervision improves contact-rich humanoid behavior beyond tactile input alone, whether the adaptation transfers across policy backbones, and how pooling, predictive context, and latent targets affect performance.

A. Experimental Setup

Platform and data. We use a Unitree G1 humanoid equipped with distributed vest and arm tactile sensors and the SONIC whole-body controller [9]. The policy observation comprises multi-view RGB images, a language instruction, proprioceptive state, and, for tactile-enabled variants, the tactile stream. We collect 50 demonstrations for each of five tasks. Within each task, all compared methods use the same demonstration set, action targets, and downstream control interface. Uni-VLaT uses the same four-frame tactile history, four-step prediction horizon, and auxiliary loss weights across tasks.

Tasks and success criteria. The five tasks cover direct arm contact, support under changing load, tactile human commands, gentle human contact, and sequential loco-manipulation:

- **Table Sweeping.** A line divides the tabletop into left and right halves. The robot uses its arm to sweep all five objects from the right half to the left half. Table height varies across rollouts; success requires transferring all five objects to the left half.
- **Box Holding.** The robot supports a box while a person adds four objects. Success requires maintaining the hold without dropping the box or its contents throughout loading, and subsequently allowing the person to take the box away rather than holding it rigidly.
- **Back-Tap Walking.** A person taps the robot’s back to trigger forward walking. Success requires responding to the tap and reaching the designated destination.
- **Human-Robot Hugging.** The robot embraces a person, with participants of different body sizes across rollouts. Success requires completing the encircling motion with gentle contact and without harming the participant.
- **Composed Cleanup.** The robot first sweeps tabletop rubbish into a cardboard box, then lifts and carries the box to a waste bin and deposits the entire box together with its contents into the bin. Success requires completing the entire sequence.

Evaluation. Each method–task configuration is evaluated over 20 real-robot rollouts. We report task success rate as the fraction of successful rollouts. The main comparison uses

Isaac-GR00T for all four methods and all five tasks. The backbone study covers three backbones and two tasks, and the ablations examine individual design choices with Isaac-GR00T. Quantitative results remain pending.

B. Task Performance and Contact Analysis

The four methods distinguish access to touch from the scope of predictive supervision. *No Tactile* uses vision, language, and proprioception without tactile input or auxiliary predictive losses. *Tactile Input* adds tactile observations while retaining only the action-learning objective. *Tactile Prediction* additionally applies the tactile latent-prediction loss, with $\lambda_P = \lambda_V = 0$. **Uni-VLaT** uses all three tactile, proprioceptive, and visual JEPA losses. All methods retain action supervision; the latter three use the same tactile input pathway, so the predictive comparisons vary the auxiliary objectives.

Table I reports the five-task, four-method comparison. Table-height changes in Sweeping and participant-size changes in Hugging vary the contact geometry, while Box Holding tests support during loading and release to a person. Composed Cleanup evaluates whether contact-rich behaviors can be completed in sequence.

Qualitative interaction analysis. The qualitative analysis will compare No Tactile and Uni-VLaT at corresponding task phases. For Box Holding, the sequence will cover support, addition of objects, and removal of the box by the person. For Human-Robot Hugging, it will show approach, arm encirclement, and sustained gentle contact with participants of different body sizes. These sequences will complement success rates with visible interaction behavior.

Contact dynamics during loading and release. Box Holding provides a sequence of load changes as a person adds four objects, followed by release when the person removes the box. The planned analysis will align aggregate tactile-intensity traces with the four addition events and the final removal. The same tactile stream will be logged for analysis in both No Tactile and Uni-VLaT, even though No Tactile does not receive it as policy input. This analysis will relate task completion to contact evolution during support and release.

C. Comparison Across Policy Backbones

We compare No Tactile and Uni-VLaT across Isaac-GR00T, $\pi_{0.5}$, and DiT4DiT on Table Sweeping and Back-Tap Walking, using 20 rollouts per method-backbone-task configuration. The two tasks cover contact-guided manipulation and tactile-triggered locomotion. This study evaluates whether the tactile adaptation is useful beyond the Isaac-GR00T implementation described in Sec. IV. Each backbone retains its native action-learning and action-generation formulation.

D. Ablation of Predictive Design

We compare the full model with four variants using Isaac-GR00T on Table Sweeping. *w/o Axial Pooling* removes the circumferential reduction of the sleeve array while retaining mean pooling of the post-DiT predictive context and future tactile target slots. *Pre-DiT Multimodal* predicts the three

future modalities from visual, tactile, and proprioceptive representations before DiT. *Pre-DiT Tactile* uses only the pre-DiT tactile representations as the source for all three predictors. These two variants examine whether prediction from the contextualized post-DiT tactile states is beneficial. *Delta Targets* replaces the absolute future targets with future-minus-current latents while retaining the three target modalities and direction-and-magnitude loss. All variants retain action supervision and the four-frame tactile history. Together, these comparisons separate the effects of sleeve spatial pooling, predictive context before or after multimodal policy processing, and absolute versus difference-based future targets.

VI. CONCLUSION

We presented Uni-VLaT, a framework for adapting pre-trained robot policies with whole-body tactile sensing and multimodal future-latent supervision. Axial tactile encoding and a causal history supply contact information to the policy, while three training-only predictors supervise future tactile, proprioceptive, and visual representations from contextualized tactile states. The resulting policy combines tactile perception with pretrained action generation for whole-body execution. Our evaluation combines five contact-rich humanoid tasks with backbone comparisons and controlled ablations. The present scope is instrumented G1 interaction; transfer to other tactile layouts and embodiments remains a direction for future work.

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TABLE I

REAL-ROBOT SUCCESS RATE (%) ACROSS FIVE TASKS WITH ISAAC-GR00T. EACH ENTRY USES 20 EVALUATION ROLLOUTS; EACH TASK HAS 50 DEMONSTRATIONS. TACTILE INPUT ADDS TOUCH WITHOUT AUXILIARY LOSSES; TACTILE PREDICTION ADDS ONLY TACTILE PREDICTIVE SUPERVISION; UNI-VLaT USES TACTILE, PROPRIOCEPTIVE, AND VISUAL JEPa LOSSES. DASHES INDICATE RESULTS NOT YET AVAILABLE.

Task	No Tactile	Tactile Input	HTD Prediction	Uni-VLaT (Ours)
Back-Tap Walking	0%	90%	90%	90%
Table Sweeping	50%	60%	60%	70%
Basket Loading	30%	60%	70%	80%
Human-Robot Hugging	30%	60%	–	–
Composed Cleanup	30%	50%	50%	50%
Avg.	28%	64%	–	–

Qualitative rollout placeholder: Box Holding + Human-Robot Hugging; No Tactile vs. Uni-VLaT

Fig. 3. Qualitative comparison at corresponding task phases. Box Holding includes loading and release to a person; Hugging includes gentle encirclement of participants with different body sizes. Rollout images are pending.

Box Holding analysis placeholder
Four object-addition events and final removal
Aggregate tactile intensity over time
No Tactile vs. Uni-VLaT

Fig. 4. Planned Box Holding contact analysis. Tactile intensity is aligned with four object additions and removal of the box by the person. Measurements are pending.

Three-backbone comparison
Table Sweeping + Back-Tap Walking
Results pending

Fig. 5. Success rate of No Tactile and Uni-VLaT across Isaac-GR00T, $\pi_{0.5}$, and DiT4DiT on Table Sweeping and Back-Tap Walking, with 20 rollouts per configuration. Measurements are pending.

TABLE II

ABLATIONS ON TABLE SWEEPING WITH ISAAC-GR00T. EACH CONFIGURATION USES 20 ROLLOUTS. V, T, AND P DENOTE VISUAL, TACTILE, AND PROPRIOCEPTIVE REPRESENTATIONS. ALL VARIANTS PREDICT THREE MODALITIES; W/O AXIAL POOLING REMOVES ONLY SLEEVE AXIAL POOLING. RESULTS ARE PENDING.

Variant	Prediction source	Target	SR (%)
Full Uni-VLaT	Post-DiT T	Absolute	–
Pre-DiT Multimodal	Pre-DiT V+T+P	Absolute	–
Pre-DiT Tactile	Pre-DiT T	Absolute	–
Delta Targets	Post-DiT T	Delta	–

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